



FULL MARKS: 40
DATE:17.05.2024

- The paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

[5 x 1 = 5]

a) $\{m: m < 2, m \in \mathbb{N}\}$ b) $\{m: m \leq 2, m \in \mathbb{R}\}$
c) $\{m: m > 2, m \in \mathbb{W}\}$ d) $\{m: m \geq 2, m \in \mathbb{R}\}$

a) $\{-3, -2, -1, 0, 1, 2\}$ b) $\{x : 3 \leq x < -3, x \in \mathbb{R}\}$
c) $\{x : -3 < x \leq 3, x \in \mathbb{R}\}$ d) $\{x : -3 \leq x < 3, x \in \mathbb{R}\}$

a) ₹480 b) ₹5760 c) ₹1680 d) ₹1752

a) 9% b) 4.5% c) 18% d) 27%

Reason(R): Maturity value – P.n = $\frac{Pn(n+1)R}{2400}$. [symbols have their usual meaning]

a) (A) is true but (R) is false
 b) (A) is false but (R) is true
 c) Both (A) and (R) are true
 d) Both (A) and (R) are false

2. A shopkeeper marks up the price of his goods by 10% above the list price and then after a discount of 5%, GST is charged @ 18% on the good. Find the list price of the good if the consumer pays ₹1881 as CGST. Also find the final price consumer has to pay.

3. Suresh gets ₹7963.25 as maturity amount at the end of two years at the rate of 10% per annum in a Recurring Deposit Account. Find the monthly instalment. [4]

4. Solve the quadratic equation $x^2 + 3x - 3 = 0$, for x and give your answer correct to:
(i) 2 decimal places (ii) 2 significant figures [4]

5.i) Simplify: $-1\frac{2}{3} \leq \frac{x}{3} - \frac{4}{3} < \frac{1}{6} : x \in \mathbb{R}$. Graph the value of x on the real number line.
ii) Find the smallest value of x for which $2x - 5 \leq 5x + 4 < 11$ and $x \in \mathbb{I}$. [3+3]

6. Find the value of k , if the roots of the quadratic equation $(3 + 2k)x^2 + 7 = 2(3k + 1)x$ are equal. [4]

7. A two-digit positive number is such that the product of its digits is 6. If 9 is added to the number, the digits interchange their places. Find the number. [4]

8. Farhan deposits ₹2500 per month at 10% p.a. in a cumulative time deposit account. If the bank pays him ₹66250 at the time of maturity, find in years, the total time for which the account was held. [4]

9. For the following transaction of goods.

Item	Quantity	Rate in ₹	Discount	GST
Cake	3 pounds	800 per pound	NIL	18%
Cardamom	100g	500 per 100g	20%	5%
Dried fruits (mixed)	1 pair of 100g packets	200 per packet of 100g each	10%	NIL

Calculate:

i) Total taxable amount

ii) Total GST paid

ii) Total bill amount [5]